

ActInf Livestream #008.2: "Scaling active inference" (2019)

Livestream | 1:25:01

hello and welcome everyone this is the active inference live stream it is act INF live stream 8.2 and it is November 17th 2020 so welcome everyone listeners and participants welcome to teamc everyone we are an experiment in online team communication learning and practice related to active inference you can find us at our website our Twitter our email on our public keybase team or on YouTube this is a recorded and an archived live stream so please do provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here in learning about these questions and as far as video etiquette for live streams goes just remember to mute if there's noise in the background and raise your hands so we can hear from everyone on the stack use respectful speech Behavior Etc ET today in exciting act imp stream 8.2 we are going to have introductions and warm-ups and then we will at whatever rate makes sense welcome Shannon we will walk through the sections of 8.2 as with 8.0 and 8.1 we are going to be discussing the paper scaling active inference by chance at all 2019 and Alec thanks so much for coming on to the show today we're going to talk about the goals and the road map of the paper hopefully from the author's perspective we'll then have a little pause just for an open Q&A because there's a lot of different directions

this paper can take one so we'll just pause there and then we'll continue through the figures just two of them and just address a few different domains of follow-up questions about the paper and then if we want to slash still have time we can go through some of the notation and math again for the rest of 2020 we are going to be moving into papers 9 10 and 11 so check it out on Twitter paper nine is going to be about Consciousness paper 10 is about scripts and about social interactions and paper 11 is about um modeling under the sophisticated effective inference framework for the intros and warm-ups if everyone could just introduce their name and their location whatever else they'd like to State especially our firsttime guests and then just pass to somebody who has not spoken so all start I'm Daniel I'm in California and I will pass it to Alec hey everyone yeah I'm Alec chance I'm based in Brighton um final year PhD student at sussex and thanks for having me on the stream we will go to um Sasha hi uh my name is Sasha I'm also a graduate student and I'm based out of David California I will pass it to Sten Stephen you're unmuted but um it doesn't have any sound maybe just reload and temporarily let's just go to Shannon hi guys I'm Shannon I'm based at the University California in Merced but currently in South Dakota

I'll pass it to

Blue hi I'm blue Knight I am an

independent research consultant and I am
based out of New Mexico I will pass it
to

Alex hi my name is Alex I'm in Moscow
Russia and I'm a researcher and systems
management school and they pass it to

Ian hi my name is

isan I'm in Russia Moscow and I pass it
to uh Steven Steven are you here I'm
gonna try again does that work yes yep
okay hi I'm Stephen I'm based in Toronto
I'm actually studying a practice-based
PhD through Canter Christ Church
University in the UK and I'm pleased to
be

here hi I'm Mel Andrews um and I'm in
Cincinnati doing a

PhD cool lot of different stages and
areas so it will be a great discussion
to bring to the technical side with this
paper for the warm-up questions people
can just raise their hand as they'd like
to speak and I will just put up the
first two questions they are for today's
discussion what is something that you
are excited about and then the second
question is what is a question you're
wondering about

so while people are thinking about it or
raising their hand one question I was
wondering about was how do we trade off
between explore and exploit in the two
figures we saw examples of explore or
exploit but what is required to enact
something that is able to mediate
between the two of them or find a

compromise between those two
strategies anyone have any thoughts or
want to raise their hand
Stephen yeah I'm also excited about the
explore um and the exploit side as well
to be honest because I think that opens
up um a big shift in the way that we
socially interact and the way that we
organize ourselves so trying to look at
a real foundational way to sort of
approach that is going to be quite
helpful I think in a lot of fields of
practice cool Alec and then anyone else
who raises their
hand yeah am I unmuted or muted sound
good sound good yeah uh I just wanted to
comment on your comment um because I I'm
also very interested in that but I guess
I'll go on the record saying that I'm
not not convinced that active inference
provides sometimes it said to provide a
solution for the explore explo dilemma I
I don't think it does uh I think it re
recasts it so you know now it's not a
question of
balancing exploration exploitation but
they're both zoomed into this different
objective function but in
practice
uh there there isn't a sort of magic
balancing act that you you get out of
this um so I think it's a good way to
recast the exploration expectation D but
certainly not a solution to the problem
awesome
blue so I kind of have like a tangential
question I and thank you Alec for your
comments because it just plays right

into I think what's missing very clear
that um if the environment is
homogeneous there's no real need for
exploration because the agent is not
going to gain any you know any
additional progress by exploring the
environment but I I just wonder where
autonomy factors in right like so you
know some agents have more or less
control over their environment and
there's no like real room for this or
not that I've seen anyway in in the um
active inference framework like where
the differing levels of autonomy of the
agent play into the need to explore
versus

exploit cool good question

Stephen yeah I think that that point
about how to look at the agent starts to
bring in the enactive component a lot
more I think and the generative model
that's kind of sitting inside the person
and existential kind of sense making so
I think that's that also yeah it's like
pushing this back from a a real world
perspective and saying okay maybe we
need to work back the other way from
complex multisensory experiences towards
active inference as well as using active
inference to do modeling of maybe kind
of very clean sort of low
dimensional approaches it's like should
we also be coming back from the other
direction um and I did I've got an
activity called conceptual action
sociometry an activity where people move
around space to understand how their
orientation is during a moment of

encounter and I've asked people to unpack it in terms of the three different areas of active influence the sort of pragmatic gain epistemic foraging or or risk mitigation or salience and it's actually really interesting when they break it down that way and start to think about what was I trying to do but from that kind of high dimensional kind of synthesis of being in a really immersive moment and they it actually people seem to who don't know anything about active inference actually were quite intuitively seeing quite a lot of shifts in those three different areas of focus depending on the type of um event or moment that they were reflecting on because they were going back to an event and thinking about the focus of their attention at that particular moment in time and how much were they sort of bringing these three areas in and then I suppose between that you could infer something about the exploration sort of desire to learn or the exploitation gain but I I agree I think it does depend on the moment that's in question so yeah I'm there's a few things that kind of this is sort of tapping into which I'm really interested in cool and to tie that just together with a recast idea from Alec it's um kind of like nature nurture explore exploit people don't know about these bin Aries they're not natural kinds until they're taught them and then the resolution to these kinds of dichotomies or binaries false or not is not some

parameter that varies from zero to one
that trades off for example when Evelyn
Fox Keller was moving beyond nature and
nurture it's the Mirage of a space
between it's about reconsidering the
situation so that there doesn't have to
be something like a bipartition so
that's definitely the direction that we
will try to take it because it's a
really cool idea the second or I guess
the last warm-up question is is and On a
related note what would be an
interesting control system or any system
to model with active inference so we're
going to go into like a trolley car and
a um Hopper I guess and a pendulum and I
think it will be fun to hear about those
but um we heard about this the social
and the spatial there's questions that
we've raised about biology communication
anything else that somebody wants to
bring up or whether it's yeah Mel please
and then Sasha
yeah um so I'm what I'm excited about
I'm really excited
about uh this kind of work um as I'm a
I'm a philosopher but um with this sort
of stuff the AC inference fup stuff I'm
like please stop theorizing just like
let's let's play with
this um so I'm really excited about this
kind of work and um the question that's
motivating me is is really sort of
figuring out um
what the boundaries of these various
models are so so what the boundaries of
of active inance are relative to the
free energy principle and then relative

to EG predictive coding models and that
sort of thing

um someone brought up recently on
Twitter I guess uh it there's a there's
a biologist at my University um who's
doing really kind of groundbreaking work
in in in like perception cognition um
perceptual cognition and stuff um
in I guess butterflies but mostly uh
jumping spiders Sal um so the the
mousse lab Nate more house is doing for
example um salti had like very uh
elaborate like um mating dances and
things and and they're signaling to each
other um and they're doing stuff like
like the evolution of color perception
in in these jumping spiders um it's
really fascinating work and they
mentioned recently um or Nate mentioned
recently that um that the lab is doing
some like simulation work on on how we
evolve additional color per set um so so
you can right you can you can if you
have uh certain like genes and proteins
for for um pursuing certain colors you
can you can kind of like shift the
probability distribution the the the range
of of color that those pick up right um
that's something that that Evolution can
kind
of modulate but um the evolution of an
entirely different um an entirely new
color percept is is something different
um and
so they're doing some some like in
silico work on that and I I thought yeah
that's what that's what active INF
that's exactly what active INF should be

do that's what we should be doing it's great so I think that's what we should be striving for thanks for that very interesting summary Sasha and then anyone else um thank you Mel for bringing up the jumping spiders um they have incredible eyes you should all look it up um but I uh I'm interested in this um from a developmental perspective and what active inference can contribute to um how systems develop and communicate and specifically uh humans um how our Ino environment and the activity um of that process um shapes the development of the human um and kind of on a more broad scale um how this can play into teaching and how active inference can be applied in the classroom at different educational levels but in a way that um sets up the environment such that agents are uh encouraged to explore instead of being dragged through the material as uh students often are so that's that's my curiosity in this time

theot cool yes lot of otsky fans slash colleagues through through time on this discussion any other comments but really great points there about like the evolution and the development of perception and then at the level that we can at least experience or enact agency can we choose to see a color as different can you look at that ambiguous pattern and choose to see it one way or another can we understand uh many of these perceptive

questions as being related to maybe not simply explore exploit maybe having an element of multiscale agency I think there a lot of ways that that could play out in people's life while also recognizing that we're not talking about just metaphors here or narratives we're also talking about this specific framework with the paper that we're going to discuss now so let's talk about those details but then maybe you know we can dip back to the other systems but let's think about how this paper and also our special time with Alec will help us understand these broader questions at least where we want to go after this short discussion so the paper scaling active inference we read the goal last time um Alec maybe if you could just however you want to or however long just what what were you setting out to do or how did the collaboration come to be what were you asking before how did you get to this paper and this goal um so I guess historically the sort of historic uh lead up to the paper was just very interested in reinforcement learning machine learning and also very interested in active inference so there a those sort of uh a natural overlap but in terms of the motivation I think the two communities don't really speak to each other as much as they should and so much of the work that's happening in reinforcement learning um especially modelbased reinforcement learning has direct analogies to the

work that's being developed in the active inference Community um so stuff like the of variational methods variational Auto encoders the use of Dynamics models the use of trajectory based planning uh or just planning in general and these algorithms have a lot of similarities between the two um the use of intrinsic objectives these Information Gain terms is widespread and both the use of belief based planning so all the things that we sort of touched on in the paper so in my mind when I started out the the aim was really just to construct an agent using the Machinery that would be common and um relatable to the machine learning crowd but that was consistent with the active influence very much um and I guess a secondary goal was also to put together an initial attempt to look at whether what needs to be put in to get some of these ideas to scale um uh with a particular focus on you know this wasn't a biologically plausible suggestion but looking at some of the objective functions the expected free energy at scale um I think is of interest rather than you know where it's commonly looked at in these smaller scet uh TM sty environment cool really helpful in a lot of terms that we've heard but to link together the ideas of the variational inference with the belief and the trajectory based planning those are all really cool things I just have a

question about the at scale part so one aspect was that you introduced The Continuous State space as opposed to the discrete but is there anything else that's meant by at scale or what what does it mean to scale in this context well I mean so I guess the scale just means is almost defined in relation to what is currently uh common in the active influence literature which is where your entire world is often designed by four or five states and there's transitions for so this paper um presents a framework that can be applied to you know observations of um you know 10,000 100,000 uh and so your state space or your observation space is much larger um so that's what I meant by by scale is is putting forward a framework that could be applied to high dimensional tasks and the ones that we currently consider and there's also a second thing which is not really captured in the word scale but also the complexity of the Dynamics um so some of these tasks actually have quite a small state space you know four or five but the complexity of their Dynamics requires um large models of a number of parameters to them um so there's kind of a scaling aspect and a scaling in in the space of complexity of cool very interesting stuff and at any point people can just raise their hand but I think we can keep on walking

through these areas and last time we did walk through the different areas in the road map just how we introduce active inference and the current state and then walk through the specifics of the model show some proof of concept experiments relate to previous work and then discuss and conclude with some of these sentiments related to what Alec is saying and I also found it interesting there's the scaling dimensionality then there's the introduction of continuous State spaces not just discrete and then often as machine learning people might use the word scaling to mean it's about the number of observations the size of the data set um and that relates to like the compute scaling relationship as you add uh when you double the data set input is it the same amount of training time is it twice as long is it four times as long is it 4,000 times as long and um so really interesting because there's also the scaling in terms of the understanding in the world world and people talk about scaling in innovation in entrepreneurship and how do you scale a solution so there's a lot of parallel meanings and a fun and concise title because uh you know that the finding isn't going to be in the title there isn't just a simple way that says oh this protein does this in this species we're talking about developments in a modeling framework but we also want to be specific about what those advances are and how they relate to systems so maybe just on the road map Alec or

anyone else just anything to say about
it or why was it arranged this way
personally I was just kind of wondering
why the previous work is at the end is
that a difference in our um academic
conventions or is that just because it
was more results
oriented yeah I mean you don't get a lot
of leeway in machine learning sty papers
um pre I mean previous work can I go
after the introduction or uh at the end
but it's always you know introduction
some methods of presenting a model
normally quite short test when the
experiments previous section very short
discussion that's kind of stand across
the field but um the reason I think I
chose prev the end was just because I
discussed active inference which
naturally leads in from the introduction
and that introduces a little bit of mass
which then naturally flows into the mod
like a previous work might have broken
up that
flow a little bit but more L cool um
makes sense um I think we can
walk through the next sections so right
now we're in the just open pause for
questions I have a few written down just
as far as just big questions because
we're trying to be a bridge in this
discussion and through our conversation
between those who might be hearing about
active inference for the first time but
they're coming from machine learning
thinking about how cool methods perhaps
with philosophical implications might be
utilized and there's also a lot of

people who are in the active inference side of things or the in activism side or the philosophy side where the details of the machine learning might be their first time hearing about it so how will we just come from where we are and think about what questions and what kinds of fun things to talk about will make that bridge Stephen first and then anyone who wants

to yeah I'm I'm just curious because I know that the scaling question has been maybe looked at more from a philosophical perspective in the kind of active inference so what this paper really helped me do or helped me was it like broke that fear of questioning the kind of these models because the like people like yourself and um annel and that are saying look there's a challenge here in the scaling like this stuff is being kind of used at these low-dimensional spaces and basically it showed like a gap in the literature in terms of how to action that which kind of helped me because it sort of gave me a bit of permission to say okay it's okay there's a gap there and it's not that I just don't understand everything it's just not easy to scale and I think this paper was the first one to explicitly say that because all the other papers tended to talk about yes we know how to scale but in a philosophical way but this is like okay well this is the problem from a a practice perspective cool just one note on that if anyone else want raise their hand I

really love this idea where there's the Gap in the literature and there's there's a billion gaps on the literature um it's about which ones are Salient and fundable and relevant so there's many gaps in literature and then when we find that gap which is often very Niche like in a PhD and I thought wow this data set has been collected from this ant species but not this other ant species there's a gap in literature we don't have this data set on this ant species and then seeing that as a opportunity and something that's okay even if it's not your additional field of study it's okay because you're at the edge like at the Gap with us talking to the author and talking through the equations but this is what it looks like to look across the Gap and try to connect because somebody else has identified it as a research question or maybe you have

Alec yeah I just wanted to comment on that as well and entirely agree with what you're saying um to come to maybe the defense of uh the existing literature on Act I think it's viewed a little bit or this maybe is just me interpreting what they're saying but um that maybe that the problems that they're um testing their agents in for instance something like the teamas are actually viewed uh as quite complex in a different dimensions so a lot of these reinforcement learning tasks stuff like Atari uh most of Atari most of you know stuff where you're already pushing to get state-of-the-art results with huge

amount of compute and Deep Mind competing against there they're often you can often get by with just this model free reinforcement learning that doesn't require any notion of beliefs or any notion of proper epistemics you can get away with under exposition so what I think uh the active influence Community has tried to do is look at simple tasks such as the teamas that actually well you can solve them with beliefs but are far far more amable to being solved with a belief based scheme and with this directed uh expiration and uncertainty reduction um to show the kind of complexity that the framework can come up but then you could also respond by saying that this is also thought about a lot in reinforcement learning and uh there is the question obviously of whether the kind of uh process varies they put forward will apply when you've got uh when you've really got the scale that reinforcement learning TS to do yeah let let me add on to that because it's a really great point so a lot of times what's perceived as Cutting Edge or modern or Advanced research in machine learning I mean go check go check the lay media and it's going to be about this many graphical processors or this size of data set or this accuracy or this level of skill using massive data and it's very much on the performance end and on the eing out a little bit more performance from increasingly large data sets with increasingly um large types of models um

but but it's actually in some sense a local exploration it's locally exploring certain Frameworks and ways of doing machine learning which is just computer statistics and so what this is like with this paper is a return to Simplicity and it's a return to a slightly different way of conceptualizing some of the parameters and how they're related and the Big Data just train it bigger train it uh better is kind of like thinking that we can do X without a belief we can train on go just by watching go and we can train the L of physics just by watching physics we can learn language by just watching regularities in human language these approaches um there's Merit to them this isn't just about one way being better it's just that what is being done in this paper is to take the free energy principle and AC of inference which previously had made these kinds of philosophically at the very least tantalizing claims like the relevance of a belief guided trajectory based optimization and search taking these very fascinating ideas a little bit out of the sandbox into the next level of the playground where now we can actually start to compete or at least compare and contrast directly with the kinds of benchmarking algorithms that are used so maybe one day you know active inference go or active inference chess but today we have the simple control theory parameters which is one level closer to these kinds of use cases that are happening today one level

closer than the te-as or the the
three-state uh decision is it a mouse or
a hawk or a cat that type of
stuff cool just the questions that I put
up just thinking about how can we
continue to deepen our understanding
which we'll move on from but just like
what's something you wondered about just
always stick with that and anyone can
raise your hand well what's something
you learned about while studying the
paper whether it was something that they
wrote specifically or whether it was
something that you kind of went down a
rabbit hole and started studying about
definitely that happened for me and then
lastly like what's something your
motivating to do more or um learn about
now that you've read this paper and have
this discussion yeah Mel thanks and then
anyone
else yeah I I guess I just wanted to
jump on that and say that um I'm I'm
used to seeing people sort
of try to compare um reinforcement
learning with with act entrance or or um
related approaches and it's nice to see
something more that's more of a a
synthesis there
cool yes so I think that if there's any
other thoughts feel free but we'll look
again over the experiments and then
we'll sort of work kind of in and out
about broader topics but let's try to
understand what was really done because
this also is the point of contact with
people who might be very familiar with
machine learning and optimization

control theory but for them it might be the first time hearing active inference so where's the common train stop where we're on board with the machine Learning Community they're interested in these kinds of tasks they frame it as explore exploit and as we've been talking about maybe there's like a bit of a re-imagination or a reconceptualization of this relationship explore un exploit or what of these variables that's what we want to get to at the end so let's for sure remember that but for now let's think about being in common with benchmarking different data sets and different machine learning algorithms so so maybe we could talk about explore and exploit how these communities think about these and maybe your observations about how they're different Alec in terms of first the example that you chose to highlight as explore so maybe tell us about the mountain car and then when you tell me I'll go to the figure one but like what is this what does it have to do with exploring what does the machine Learning Community think about it how does active inference apply like what is happening here yeah so Mountain car's quite an interesting one um it seems so simple it's two when you do it in a fully observed environment there's two states one action uh this is the continuous version so it's continuous actions so you know it's like the simplest task that machine learning people present in papers but it's actually one of the

hardest so um you know dqn vanilla dqn
doesn't stand a chance of solving this
what is dqn sorry
uh d q networks have to remember uh so
that's taking like the old um the old
idea of Q learning which is kind of was
the biggest idea of reinforcement
learning before this whole deep phase
and just uh amending with deep networks
it's the one that

um

be

Atari uh yeah the first like Deep Mind
nature I can't remember the details but
it was It was kind of the the birth of
the deep reinforcement moment great so
that one has challenges on here and then
just continue I mean basically the
reason is is because you only get a
reward when you reach that um flag up at
the top and before that your um you know
that takes about 180 steps or 180
actions that you have to um string
together before you get any notion of um
reward so just pure reward-based
schemes uh really struggle with this
they essentially have to do more or less
random actions until you get um out of
the Hill um so yeah it's it's a struggle
and that's why it's kind of exemplifies
exploration because you need to care
about learning about the
Dynamics um or some other intrinsic
quantity if you want a chance at finding
out how to solve this

PA cool very interesting blue this is
just really reminding me of a lot of
what we've talked about with explore

exploit a lot of the perspectives you've brought about how exploring how and when and where does it matter so pretty interesting to hear Alec just how you phrase that and also just like to hear that it's 180 sequences of actions so this is not like control theory like Connect Four this is like walking you know who knows how many degrees of freedom there are but this is many tasks that have to get strung together in Long sequences and then you said something about how um it has to learn something intrinsic for example the relationship between its velocity position and policy a little bit more nuanced Nexus of action rather than just learning like simply whether you know go means fast and stop means slow Stephen and then anyone else yeah can I just ask the question about that when you say stringing together um actions because I I kind of had this feeling that it was kind of a case of like you keep it the by by there being a desire to explore in itself through free energy that the car realized that it could go up the other side of the Hill further because it could just have a reason for going there because of Trying to minimize the desire to free energy around exploration and in doing so it gets that extra kind of momentum to get up the other side of the hill but I haven't really g into a detail so I was just wondering how this kind of sequence of steps and maybe just finding yourself far up far enough up the other Hill to get enough potential

energy just to go down and up and make
it to the flag I wonder how those two
sort of
relate great question AI maybe if you
have a thought on that yeah yeah so I
think your intuition is mostly correct
so how it plays out is that this agent
is evaluating different sequences of
actions some of those actions are going
to keep it where it's already been which
is at the bottom of this hill uh and
then some of the actions are going to
take it into a place where it's
uncertain about the
outcome um and that UNC and that
uncertainty is essentially valued by the
agent it wants to resolve it
so it's going to say okay what happens
if I do a little because it has to go
left up the hill I should have mentioned
that before it has to go left up the
hill actually actually go away from the
flag to gain momentum to around so it's
going to you know it's never gone left
up the hill and then
accelerated um doesn't know what's going
to happen when it does that that's
valuable to it in terms of epistemics
and then it realizes oh wow up the
hill yeah let me let me link that um to
the question the sequence of of
so imagine if you were going to do a
control theory optimization on shooting
a a bow and arrow so shooting a bow um
is a complex motor movement uses
probably many joints there's a lot of
you could do different speeds different
ratios it's going to be dependent on the

bow just like this is going to be dependent on the slope and the car so it's it's an enacted affordance that you're trying to develop an action sequence for and your action sequence in uh game is often just one stepper Drop The Connect Four token here I'll re-evaluate after I see what they do but if you're going to be evaluating prospectively an action sequence in the world it often has depth so it's like getting out of my chair shooting the bow there State spaces are very there's multi- um joint so it's multi uh dimensional and it's continuous so that's why the multi-dimensional and the continuous are so important and then the depth through time is so important because you can't just be doing a get out of my chair or a uh shoot the bow and arrow short-term One Step optimization there's no opponent for you to then take a look at their move like in a chess Checkers go Paradigm or even a video game Paradigm to some extent this is something about planning in an enormous State space and thinking really constructively about these intrinsic relationships so in the bow example you might be attuning to the tension between the propri reception and your shoulder and how tense the bow is to know if you're at the end of your range of movement or the bow is and if you trained on one that was half size and then you go to a larger one or if your arm is hurting that day or all these

differences differences in our abilities
these things all become enacted in the
relationships that we're learning about
the intrinsic variables so in the depth
through time in intrinsic variables if
you learn that hey if I actually reverse
and start accelerating downhill a 100
times steps later it's just better like
if I go to sleep at this time 100 times
steps later it's just better and so
these allow for very nonlinear policies
to be selected because there can be
temporal depth that's learned as a
function of the actual physics basically
of the setting not just like
counterfactual you know if 13 moves down
the row this person does this with my
Rook that still is in a very if then
context and this is taking it into a
totally different domain with a
proactive action selection
policies yep so that's why this is such
an interesting paper and approach and
like again a branching off point that's
why it's foundational in the machine
learning because just like Alec
described like if deep Q learning cannot
accomplish this task then yeah it's all
great to be a human at chess or go but
if that algorithm or even that
architecture can't defeat this challenge
then we're developing something that's
incredibly specific which is great we
should have good map routing algorithms
and things like that but we're
definitely going hyperlocal down a
rabbit hole if it can't solve this but
it can solve go I don't know if it is

that case um a machine learning person would be really welcome to help fill us in on some of these details but that's the kind of stuff that is um interesting and and broached in this topic so just to see what that looks like empirically like the first 100 epochs um are plotted in terms of the state space coverage which here is just the position on the x- axis and then the velocity on the Y and the flag is at 0. five right Alec like the Flag represents being at position of point five but yeah continue from here just anything you want to add on figure one or what what does this mean what is happening here I guess no your description great the flag's not at 0 five the flag's uh what is it oh yeah 0 five yeah uh but I just kind of want to um self deprecate myself and say that these results kind of they're not I mean they're not bad the result but um we had a follow-up paper where we got this all working properly um called reinforcement for inference um which is very similar but we use a slightly I can discuss the differences I think they're interesting if we want but uh just that the the results are so much more impressive we can still uh it's kind of both uh hitting on this and also um here's the to advertise that that work so in that work oh yeah if you can actually just show the figure and that we can sh so there it can solve the mountain car in a single trial and I have also the sort of State space spots that look a bit better so here's

the so just one yeah so if you kind of
go yeah I wish had the video it can just
the first time it goes into the ring it
just solves it straight away which is
much nicer so the other the other one in
the paper that we're looking at today um
so the here it's being solved every time
that it does it and it just con gets
out and we try some harder tasks in this
like the half G2 and ant maze coverage
gotta love

it um yeah I guess that was just a I
wanted to highlight that um it can do a
lot better than this um but I'll be
happy to discuss the differences because
it's just one part of the architecture
that sure yeah what takes us from here
to what we just sort of peaked into in
this paper like what was the one thing
that you added or changed yeah so when I
wrote the original one to um so to get
this kind of

what what Carl calls parameter uh
exploration or parameter Information
Gain it's essentially you're trying to
reduce on certainty not about what's out
there in the world but about your own
model you have beliefs about um your
model uh and it's kind of in a sense is
you know what you don't know and to get
that you have to have a
distribution over your model and in this
case our model is a neural network to
need a distribution over your neural
network there's two ways is in the
literature to do this in the first paper
scaling active inference we tried B in
your networks uh there may be more

principled you know you're actually using variation inference to estimate this um distribution of each of your parameters but uh in practice they don't work nearly as well as what we tried in the next paper uh which is called The Ensemble approach uh and the idea there is you just take in this case 2 Dynamics models and train them on different batches of the data and that's kind of like a proxy for a nonparametric Bas in posterior over the um Dynamics model um you can do a little people are kind of working out how close it is to actually proper Bas and beliefs but um you definitely get a notion of uncertainty there so here um and there's lots of interesting reasons as to why principles reasons as to why Ensemble models would work better than um bin models it's also easier in practice to estimate stuff like um information game um so that just turned out to be 100 times better and in the machine Learning Community people generally find more success more success with uh deep ensembles for both um uncertainty calibration and directed explanation that's very cool thanks for the great explanation there and so just to sort of rehash that or say it a little differently the approach that was taken in this scaling active inference paper to estimating some of these essential parameters was done with the basian neural network which Alec just

described it's like one way to do it it might be a way that interfaces pretty cleanly with a lot of other software packages or approaches but it's the skeleton of the model and that's the diagram and then we think about how can we improve it and so One Direction is more analytical or more principled as you described it and that's like the fully Bayesian sort of specifying every little uh hidden State and that approach can provide some interesting Avenues at times but an approach that is also relatively easy to implement from a programmatic interface level and also makes a relative minimum of assumptions about the specific mechanics of the system is this Ensemble or deep Ensemble approach so the Deep adjective is just meaning you're going to have like a multiscale or I mean a multi- level neural network with hidden layers or it's just going to be deep and modern but it's kind of like just an adjective The Ensemble approach is describing having a bunch of different models that are going to be trained up on the same or different parts of the data set and then you could look at the average of The Ensemble or you could look at some other weighted combination of the ensemble's performance and so it's almost like Bridging the Gap from Individual testing that's like the single model testing to Now The Ensemble approach is like we're going to have a classroom and then the best answer from R9 group all working on it independently

the best one will push forward or will average and then even another level beyond that is like the truly emergence which is actually what the ants are doing where the Ensemble works as an ensemble in a way that is itself shaped by development and learning and evolution so right now The Ensemble modeling is still like sort of well if you just split it up into many parts you can cover more State space you might be able to train the model in parallel you might make sure that no single model over generalizes there's so many benefits that come simply from batching and Ensemble modeling that go beyond just saying well what's the best single model or what's the best parameter range in this type of model The Ensemble can consist of a single type with a different parameters between the different Ensemble mates or they can be heterogeneous in some aspect so just really cool you went with what was implementable and a common Point of Departure with machine learning community and the Bayesian neural network kind of peaks in one direction towards more analytical more principled fully Bayesian approach and then on the other side to Modern approaches in machine learning like deep Ensemble learning cool let's look at figure two and what is conveyed here so this is the hopper task oh we'll go Blue first go ahead sorry I didn't see that can you back up to the last figure let's do blue and then me and then anyone else so um I

just wanted to ask and this is kind of related to the next

figure but um I just wanted to ask that uh when you did the hopper task and the um pendulum so I know that that was only based on the extrinsic value like part of the free energy equation so like you left off the exploration because like exploration probably had no value in in that problem I mean I'm assuming there but I I was wondering here like on the exploration problem sorry did I say exploration before exploit anyway so here with the exploration problem um did you ever think about leaving off uh the extrinsic part of the equation and just using like the information gain as a reward like can you structure it that way would it be different like if the object of the agent was just to gain more information about the environment do you think that it would have succeeded in um you know climbing the hill because it would get to a new part of the of the environment or how do you think that would

work yeah yeah uh not in this p but it does um so that other paper I mentioned the reinforcement through active influence because it solves it in the very first task it literally can't be because of reward we've tested it without reward as well to confirm this but you know it hasn't ever experienced a reward and it's still solves the task so it's exactly what you're specifying it's just purely trying to reach all the parts of the uh the state space uh if

you just left it with expiration what
you tend to find is that it will solve
it for say the first 10 trials while
it's still the peaks of
the Hill are the most kind of
interesting because they have the most
extreme Dynamics with the most variant
um but then after a while uh the top of
the hill is no more interesting than the
bottom of the hill because it's kind of
explored a bit whereas if you've got the
reward and the expiration kind of slowly
transfers from solving it because of
expiration to solving it because it
knows how to get
reward awesome awesome thank you I
didn't see that followup paper I like
I'm going to go read it right after this
I know we need to do it a a
Afterparty active Afterparty with next
paper um but uh Mel and then anyone
else yeah just um what you were
describing with uh like stringing a bow
something like this um I think that was
what really got me excited about active
inference in the free energy principle
in the first place is is the idea that
um with something like a like a a towers
of Hanoi style problem are people
familiar with that uhhuh like you've got
you've got three sticks and and you've
got discs of various sizes on the sticks
and you've got to get them in in a in a
like a ascending order right um and the
idea is that uh the puzzle requires that
you you go backwards before you can go
forwards and if you're just
minimizing or maximizing some function

on a single level um if you've got a
like a a ond
optimization you're you're going to not
be able to solve a problem that requires
that you backtrack in order to make
progress um and something like active
inference the free energy principle
where we've got temporal depth we've got
hierarchical depth of the model um is
really equipped to solve these kind
problems in a way that a lot of
traditional um problem solving
approaches are not I I totally agree and
I think there's a quantitative and a
qualitative so at the quantitative level
there's control policies that we can't
have computers look Beyond some locally
non-favorable states to get around so we
want to do these quantitative policies
that's what this paper is about but at
the qualitative and really the
philosophical level how do we come to
grasp with processes where yeah we're
not always strictly walking a staircase
directly to the top of the mountain or
it might not seem like we can do it at
all initially if we just look and
there's so much there with how we think
about challenges and about exploring one
actual uh link there cuz plue asked
about what would happen if you just let
it go wild on exploring and Alec what
you said is that the learning gets you
to the top pretty quickly because in
that sense it's similar to this model
that also prioritizes reward but then
after getting to the top you spend a lot
of your time learning on the most

extremely variant areas of parameter space so it's so much like curiosity driven learning where a lot of times curiosity driven learning with no reward or scaffolding it ends up learning learning a ton shocking amounts but then also spending a lot of time in the most extreme ranges of space and that doesn't always just play out like in the kind of online gutter but even in the literature we see a lot of the attention being spent on the extreme hyperboles and then the middle ground where it's like yeah it's kind of balanced and we can work on it together that is not as extreme of a Viewpoint and so people spend less learning and attentional regimes on these kinds of projects so there's a lot of like parallels that we can quantitatively model but also help us think about how we can say no you're you're just trying to make me explore the top of the hill I get it but the flag's on the other side so I've heard about the other side of the Hill how are we going to enact a policy to get us to the flag together that's a little bit better than your Hill is worse than my Hill for example how could these kinds of things be ported onto like human decision-making is a cool area any other questions on one otherwise let's talk about two a little bit well just just one quick piece just picking up on what blur said there I thought was quite interesting is if there's this idea of exploring the

environment in different ways but say there's exploring or Information Gain in terms of personal preference so for instance maybe it was a bit more of a con I like to always feel what it's like to go slightly up a slope and turn to the right that feels cool right so then you have this kind of like it's not necessarily in it's like an Information Gain about having fun so would it like say the car is like hey I really like to do this type of things because cars like me like doing that I know you know if it was a living animal so it kind of ties into what sort of exploration of the world and then What exploration of just being an entity that likes to do certain things and that could end up taking you there in another in another way or together they help great

Alec yeah I just wanted to comment on that because this is uh I totally agree and I think it's one of the most uh promising kind of directions or perspectives that active inference gives and I wrote about this in a in another paper that I can link afterwards uh of this notion of I don't really know what to call it but but goal directed expiration that you derive from something like expected free energy and that means you don't necessarily or it's just too inefficient just to have everlasting expiration with no constraints you know um uh and it's also too inefficient just to do have exploitation what you need is some

objective function like expected for energy or model evidence that contains these two things so that you're not selecting actions to just explore or just to exploit that each action is kind of shaded with both of these things and that way you're getting a a expiration that's um geared towards your goals and it greatly constrains the type of exploration you'll be do and I think from nine you know that also fits with our kind of experience maybe of of daily life or maybe not but I think from an engineering perspective that that's crucial because in real world task there's just too much to explore and you need to prioritize that exploration I think that's something yep that's so beautiful and even another layer is The Ensemble of mountain cars so now imagine we all see different things and we don't know the way to get to the top of the mountain we don't know the policy we don't even know what the end point is we don't know if the one that we can see close by is the best one or if there's a way better one way further away and then everybody is who they are and they all have their own landscapes and then through the ensemble's modeling collectively with or without information sharing of whatever kind The Ensemble as we're seeing gets better performance there isn't just one best policy of mountain car there are so many different ways to ascend the mountain and then you open it up with what the objective

functions and the policies the goals
could be and it really is a a great
space Shannon and then anyone
else hi thanks I was just um thinking
about your Ensemble of of cars here and
comparing this to like flocking
behaviors in birds or even um people who
are foraging together and maybe I was
wondering how so maybe this entire flock
is minimizing its free energy by getting
closer to award which is food but any
individual bird in the flock is just
following local rules about like how
close to fly to other agents or when to
follow them or when to break off and
find a new like go away from the rest of
the flock or drag the flock along with
them and I wonder if every single bird
is having their own little mountain car
model or if there's just one gross model
for the entire flock as a mountain car
finding finding its
food yep well one angle on the foraging
and affordances is Imagine That ensemble
of human foragers but they're not
exactly the same size or they don't have
exactly the same preference or they have
slightly different Visions some people
see closer or further they see color or
they don't see color and so all these
differences are what allow The Ensemble
to explore and exploit especially with
information sharing oh hey I was over
under this tree and I found this but
somebody else wouldn't have found it so
we can all use the Ensemble to take
advantage of our um you know differences
of the nestmates or the flock or

cognitive diversity and then in the question that you raised about is there a single uh layer being enacted by the flock or is there a little mountain car as you said by each bird so from a modeler's perspective if it turns out that we can explain variance about the bird's trajectory by putting a linear aggression on it it doesn't say it's linear aggression that the bird is doing just that it helped us explain variance in the world so similarly the low bar here is that we can use this kind of model just like we could use another type of control theory model or action policy selection model model to explain variants about the real world and especially because we see organisms succeeding in fact it's really the only ones that we do see this helps us explain those successful systems as opposed to like funny little gifts of you know robot flailing and then whether there's like one layer that's being enacted and then the group is purely epiphenomenal and there's no downward causation there's no influence of the group states on the lower level states there might be some systems like that there might be other systems where there's an analytical solution like a well-defined solution at the bird and the flock level there might be another one where it's well defined at one level but then because of interactions and emergence it isn't as defined or isn't defined by the same class of model at a higher or more coar grained

level any thoughts yeah that's all great
thanks y cool let's look at figure two
so similarly as you did for that um
Mountain car which was like such a
helpful direction to take what can you
tell us about the hopper V2 um or the
inverted pendulum which I don't have
here but those were the two TXS maybe
more on the hopper um or the pendulum
whichever one just what are they about
what Will machine learning people
recognize or know these models as
signify um so
Hopper isn't um it's not expiration
based it's quite a dense reward that you
get so you kind of get rewarded or dis
rewarded for each of your action but uh
it's slightly more High
dimensional
can't it says two dimensional there but
that's just the that's not when I speak
say Dimensions I mean how many
observations it receives um it's more
than two it might be like 16 I'm not
sure uh and it's just generally the
Dynamics so you've got to learn you know
you've got maybe four actions which are
real value numbers between minus 4 and
four you've got to learn how that
changes 16 variables over the course of
your however long you're planning so
it's it's a much harder task to learn um
so I guess that's why it was um included
in in this just that it's generally
valid is a bit of a hard task in the
machine learning commun cool and so what
is happening in figure two what can we
say how is it different than ddpg and

what actually is ddpq for

reference ddpq is uh the so acronyms and

deep deterministic policy gradients um

so what is that one doing and then what

is active inference doing that different

that maybe enables it to have such

better

performance um so what is uh ddpq doing

so it's just a it's a model 3

reinforcement learning algorithm so you

have a policy that maps from states to

actions and you essentially do a load of

maths from something like Bowman

equation to get to

um an update for your um policy

parameters um the reason that it's doing

so much better the reason that

modelbased reinforcement learning in

general does so much better um is

because it's essentially it's a planning

algorithm so

um what's the best way to describe this

yeah

um due to the fact it's model based it

can learn from every

single um bit of data that it receives

so that is in terms of each state

transition and as well as the reward

signals um whereas this model 3

reinforcement learning is is just simply

learning from that single bit of

information they gets at each time step

which is uh um the corresponding reward

or 32 bits or what however your rewards

work ex thinkd um so I mean eventually

ddpq after enough EPO would probably

asent it around or higher than um model

based that's a kind of

General uh pattern you get with model base and model 3 reinforcement learning model base is far more sample efficient uh but model 3 takes a hell of a lot more samples but is um ASM tootes gets you know levels out at a higher reward let me also add a layer there um with a few other aspects from machine learning and the idea of the ruggedness of the landscape so if you're doing this task which was described as a dense task another way of thinking about these is you're trying to keep something upright so it's like very obvious if you're succeeding or failing and in the big landscape there's really easy to tell differences between succeeding and failing that being said policy planning especially when you have four variables of control that are projecting out to like 16 potentially nonlinear connected outcomes when you're trying to do that there may be many strategic mappings there might be many policies that help you keep the pendulum up like for example going a little bit Back in Forth with a certain speed or you can go a little further with a different Rhythm so there might be a many many different ways even for a single joint of control many policy sequences Through Time many learned relationships that help you stay in that yes no area now if you have a model-free reinforcement learner it means it's learning basically the raw connection it's model-free socalled between the reward and the policy and so it may spend a lot of its time exploring

locally a policy because it's like in the spotlight it's working and then it goes to a slightly different area it's not working where do we go from outside of the spotlight it's all dark no idea model free with the Deep generative model it's not like this Spotlight in or out reinforcement it's like we're learning these intrinsic relationships which actually helps us get a grasp of the maps that are going to guide us to search on the territory a little bit more exhaustively but then that part at the end which was so interesting that the model free sometimes gets to higher final performance in some context that can be because a truly model-free search can result in some wacky combination that wouldn't have necessarily been approached that it just uniquely potentially not in a resilient way but it uniquely allows some performance on a task and so that reminds me of like the evolutionary computation where its goal will be to travel distance and then some will actually go down the walking Road and those start slow but they could walk forever and then other ones just like fall and so it's kind of like hacking to get over um a barrier without necessarily uh a deeper understanding of the ecology because it's so blindly pursuing just performance so I know there's a lot of stuff there but this is like really an illustrative example and it does highlight a lot of the differences between the model free performance and the state-of-the-art in

that as well as what potentially active inference could

bring any thoughts on to now we'll just have some general areas basically the first area was implications I know we've probably talked about some of them but three areas that I thought about were like robotics resources and allocation and then rugged Landscapes this one probably in the southwest um so any thoughts on these I think we've almost touched on few related ideas but if anyone wants to like speak to one of these areas of possible implication whether they do they don't see what an implication could be or what would another domain of implication be um yeah

Alec go

ahead so

um I also am becoming interested and so are some other people in um whether some so I'm very invested in this idea of kind of a Bas and brain and um as active inference kind of as FES but whether some of these ideas from Bas and machine learning that are used in this paper so stuff like amortization um might also be employed in nervous systems um so you know we've had a few very very uh big proposals for how the brain might Implement Bing infl stuff like population coding predictor coding uh the processor that's most commonly associated with fact of inference in this terms of message passing but it might also be uh you know this amortization is also um another real possibility of how the brain

implements some type of influence so I think it could have implications for understanding neuroscience. I did not expect that message passing and predictive processing predictive coding type models can be understood alongside amortization models as an alternate implementation or mechanism of Bayesian brain as a specific testable hypothesis so this is where the rubber hits the road with the modeling and with showing the Deep mathematical isomorphisms between these different kinds of relationships like there might be some paper I'm sure there is where like message passing is equivalent to Bayesian networks and then that allows us to bridge two big areas of literature and so if we do a lot of investigation in the brain and then we find out wait there's actually multiple ways that that could be implemented in different systems whether that's through message passing so maybe that's more applicable to a computer network maybe it's also going to be implemented by policy planning ensembles so then whoa what else is doing this kind of Bayesian like processing so there's so many cool directions there great topic because also robotics or at least the question of implementation of action and selection of policy and from sensors with actuators resources in terms of informational attentional whatever they may be and then rugged Landscapes is just that's everything that is policy selection under uncertainty for any kind

of system that wants to stay alive let's
go oh sorry Sasha or blue oh see you
later blue oh wait Sasha out blue please
thank

you wait muted but yeah go ahead uh so I
just I had the same reaction that you
did like basing brain amortization like
wow I just was wondering uh Alec if you
could maybe just unpack that like in a
very simple way or elaborate more on on
that yeah uh sure so

um the best way so I mean amortization
um as it's realized in stuff like
variational cers but also other schemes
um we can like maybe discuss a bit more
to get what defines amortization are
just an instance a way of learning a g
model and well more precisely doing
inference uh the key defining features
for me although I don't think this is
ever been properly defined in the
literature is this notion of having an
encoder um how that would play out in
the

brain um how it could play out in the
brain is just kind of a um a feed
forward mapping um feed forward part of
say the visual cortex that maps from
some data or a lower part in the
hierarchy to some posterior parameters
higher up uh in the hierarchy and when
you have stuff like um population coding
or predictive coding there's not this
notion of quickly in a fe forward manner
mapping from data to parameters you've
got this kind of iterative procedure
that slowly um and um sequentially
updates the beliefs those posterior

beliefs based on the data based on some learning so it's another um route towards B INF it's far far more fast and doesn't require recurrent processing and that kind of speaks well with some things we know about the brain it's not all the brain the brain definitely does have Rec of processing but a lot can be done in in a kind of Fe forward manner very quickly uh and and the second thing is that um the parameters that you're optimizing of the encoder are optimized um over the entire data set whereas in some like predictive coding it's optimized individually the posterior parameters um of your beliefs optimized uh based on the current data point uh and I think that has some that might speak to some of the generalization capabilities of Bas influence um and that's something that we're looking at at moment so more obvious I hope that kind of answers your question R bit no definitely thank you so much totally epic to connect that feed forward encoder model to the machine learning um side so so instead of doing like a back and forth expectation maximization where you're um just updating back and forth or especially a uh like a back propagation type thing where there's very very complex interactions about how parameters are trained this allows us to kind of train on the Fly and just learn as we go now to the second point which is the usage of the entire data set versus Point by point this is like we're

learning on the fly but um we're
learning on the fly in the state space
that we want to be learning about with
respect to the entire data set that we
have access to which is a little bit
different than like you're kind of
trailing your finger along a Time series
and then you're updating based upon that
data point and what you believe and what
you've recently seen that's like a kind
of Point by point way to use a whole data
set to update parameters versus this
variational approach which actually
enables like almost a simultaneous
utilization of the entire State space
I'm not sure if that's totally correct
but those are just two parts that hit me
about the two sides that you mentioned
Alec Stephen and then anyone
else so I've not heard of this
amortization so it's quite interesting
it's it's almost like an reverse then is
it it's like you you'd have a a big
complex space of knowing and you quickly
discount stuff and the process is about
discounting away rather than building
up the model is that is that kind of the
the
idea uh I'm not sure if I've understand
understood exactly what you're at but
for my from I'm not sure if it if it is
but maybe we could get into that a bit
more how would you define Alec how would
you define amortized inference like we
see it here in the paper but how would
you define it you mentioned some
benefits we've talked about them a
little bit but like in this kind of new

way of thinking about it what what does it mean or what would it mean for the brain to do it or what happened programmatically I mean so this is what I was getting earlier it doesn't really have a clear definition um I I try and Define in two ways one is how it's generally used if you see amortized inference in um machine learning uh you're generally you should think of a encoder Network so you've got a neural network it takes in your data and it will output literal values for your parameters that would normally and those parameters are the ones that would normally be optimized in traditional variational methods here it's just um they're just spat out by the network um and then you have some other um learning method which which updates that en code for you and also your generative model but um in terms of what amortization actually means going back to the word I think derives from economics um I think it's this idea that you come to the second point that I made where uh you're sharing the parameters of that encoder across all of your data so you're amortizing the cost of inference um is how I think about it I could be I could be very wrong um so that just means you you kind of don't reset with um each new observation or you're not trying to optimize with respect to um each new observation and then kind of discarding the information you're just sharing that encoder um the the encoder which is the relationship

between the data and the parameters
across uh your entire lifetime so you're
amortizing the cost of infant is how I
think about it so a better
quote so is it like is it like you so as
opposed to just discounting data it's
discounting ways of chunking up the data
or ways of analyzing it if that makes
sense like you could have lots of
different models and they say okay I'm
going to apply them in a more efficient
way over time as you start infer it's
almost inferring what models should be
deployed is that the kind of idea and
you start to take away the models which
are less useful in a way I guess you
could another kind
of people sometimes refer to it as
learning to infer so in which might be
what you're getting at um so in normal
inference you're just doing inference
but here you're
learning how to do inference so the
learning um yeah so maybe that was what
you're getting up with your kind of mod
selection tape cool thanks yeah that
that that makes sense I don't know but
I'm trying to find my way through
there's a few Dimensions so it's really
helpful to unpack it there's the
learning how to learn there's that meta
learning element but also this is really
important that in the paper is the
number of parameters remains constant
with respect to the size of the data so
if I'm trying to estimate the mean and
the variance of a normal distribution I
know that I want to squeeze down

whatever data set I have into two numbers and so if I have three numbers coming in the pipeline I still want a mean and a variance if I want a billion numbers to come in the pipeline I still I want mean variance and so it turns out that by knowing and by specifying ahead of time that you know exactly the size of the outcome it allows you to uh scale a lot better because you can know that no matter what you're going to put in on the inside it's going to be able to be distilled down very rapidly whereas another method that results okay well we put a data set and then we find out how many principal components explain it best that might computationally scale in a way that is very um disadvantageous with respect to the size of the data whereas this method can um do that and then as a sort of corer there it's happening through a single forward pass of a network which can be AB initio sort of like denovo setting it up from the beginning and or updated in a learning fashion and so there's there's a few dimensions and Alec really look forward to you know seeing what you continue to do with the amortized inference seeing as that concept becomes a bit more formalized because this seems like really powerful and then if it also turns out that it's an implementation of basian Statistics or basian brain then it just Taps into the entire message passing compute graph um basian NET Framework which already has been the um

most helpful for machine learning in a
lot of other areas too
very very cool
so I guess in the next steps which is
really I guess the last thing that we'll
talk about for the last few minutes just
thought of a few areas to go into for
Next Step so Alec definitely you take
the first pass but the areas could be
like computationally what does that mean
Hardware software from a math
perspective what could be analytically
shown or what relationships would be
good to know um from an applications
perspective what robots need this kind
of software update
and then just educationally how do we
come to you know control our attention
so that we understand these Concepts as
well or what did you start working on
after this paper I guess it was yeah no
worries no worries I was I wasn't yeah
but I guess you did this one or where
are you at now steps are um I guess on
this line of thinking that the um this
line of work the stuff that I'm still
thinking about is what was mentioned
right at the beginning which is the
balance of uh exploitation and
exploration and whether the active
influence perspective provides you um
anything in
practice um because you know we've got
this nice objective functional expected
for energy uh and we can optimize
actions with respect to it and it does
contain exploration exploitation and it
motivates exploration and exploitation

from first phasing principles but in practice you're going to be in any model you're going to be uh fine-tuning those um the the balance between those um you've got a bit more of leeway to do it because now you can um change the shape of your PRI beliefs for instance uh and change a few other parameters that are going to lead to that payoff rather than just having some scal of weight that defines them so some of the stuff that I've been looking at is yeah learning prior beliefs in order to facilitate exploration exploitation um because I think the the to summarize that yeah the balance between exploration exploitation is is probably one of the big unsolved questions in um control reinforcement learning Neuroscience Etc um and for making these machines work um as well as we'd like them to uh and active influence offers a new route to try to solve that I don't think it has been solved awesome that's really exciting there's just so many aspects to the control theory question when there's a lot of solutions that could work and maybe it's even unclear what it would look like to work especially as we think about some of these systems that go beyond um the merely just keep a pendulum standing but what if it's give someone a massage well that's a little bit of a different question it's relational or how are you going to exercise in a way that's comfortable for you these are control theory questions that are literally

about action sequences and so when we start thinking about how we're going to apply it to different systems having a framework that is at least moving in this direction of what you said is like with the reconsidering the explore exploit by adding things like reconsidering the shape of your prior so how can you step into a framework where the explore exploit is a dimension that could be calculated or described or kind of summarized but isn't simply the underlying framework of the model we're going to get exploratory Behavior we're going to get so-called exploitative or narrowly searching behavior that is not being ruled out by what we're discussing we're talking about another way that optimization could proceed that doesn't use simply the explore versus exploit um or the coefficient weighting of these two through time or statically to um outline its whole learning approach so Stephen first and then anyone else if they want to give any like last thoughts but this has been a great conversation so even than anyone else I mean also if you go into the real world environment you've got this explore exploit situation but in high stake situations you've also got um like they often talk about risk mitigation versus gain optimization if you're in a kind of um a conflict or you know a legal situation because it may be that it's a question of you know which is in a way gain optimization is a bit like an explo exp it but it depends which way

you're going it could be an information game but whatever you're you're trying to do those two two aspects sort of come into play depending on the state of the of the situation you know so you know explore exploit is less relevant when you're standing on the edge of a cliff playing a game you know because it's like there's a risk mitigation issue so anyway I thought that might be interesting I just thought about you know doing learning action policy but if you're learning an action policy where you could die if you're rock climbing or something it's going to change how you learn and so when you're trying to minimize risk versus failure attempt or minimize the number of opportunities you have to see it successfully performed or you can only infer it by observing it these things may contain a ton of information in terms of how real systems learn where failure is not an option Mel and then anyone else who wants to close it

out yeah sorry if this is I guess a a sort of an ignorant question but I I thought some of like the beauty of these act inference and FP approaches is is that they're sort of doing a a kind of like multi-level Dynamic aoms razor type thing so they're they're maximizing predictability well well also kind of minimizing the complexity of of what they're doing

and so I guess what I don't quite understand is how the amortization of inference process works but if you're

fixing the params doesn't that kind of
uh constrain the ability of of these
approaches to to do that kind of arams
Razor type type
uh um not really because the parameters
of your encoder aren't the parameters
that you're they're not the free energy
parameters they're mapping to the free
energy parameters so you could map you
know to uh you know fre just says that
you've got to maximize yeah the accuracy
or the likelihood while minimizing um
the complexity so as long as your
encoder maps to the part of Lea stat
that is maximally uh accurate and also
minimally complex then you're still
minimizing um complexity while
maximizing accuracy uh and then
obviously there's what we haven't
discussed is the learning schemes for so
the way you learn your encoded
parameters is so that they out but
something that does conform to minimal
variation yes okay so you're not
actually
constraining you're not actually placing
like additional constraints on on the
parameter of the actual not on the
parameters of the encoder on the output
of the encoder
yeah I guess that's a good point uh so
the the encod the I guess a problem of
armatization is that um the encoder is
kind of not part you shouldn't think of
it as part of your generative model it's
just almost like a a tool that can
um map to the belief space of your
generative model

um yeah just
so just on a closing note there that
really returns us this dual
instrumentalism with rappers and rappers
and is it what the system is doing or is
it how we're looking at it these
questions are very rich and so it's a
great conversation really thanks
everyone for participating this was such
a helpful discussion and I think
definitely while R listening we'll all
pick out some questions for ourselves to
follow up on and some curious things to
learn about because there are so many
good ideas brought up so if people who
are on live if they check their calendar
event they will see a feedback form
which would be helpful and anyone else
who's listening or watching please
provide us with feedback suggestions or
questions but other than that just um
stay in touch and everyone awesome work
for this helpful discussion and we'll
see you soon